Water Research 166 (2019) 115080

Contents lists available at ScienceDirect

Water Research

journal homepage: www.elsevier.com/locate/watres

Intermittent micro-aeration: New strategy to control volatile fatty acid accumulation in high organic loading anaerobic digestion

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ARTICLE INFO

Article history:
Received 6 June 2019
Received in revised form
2 September 2019
Accepted 10 September 2019
Available online 11 September 2019

Keywords:
Anaerobic digestion
VFA accumulation
Micro-aeration
Oxidation-reduction potential (ORP)
Microbial community
Metabolic pathway

ABSTRACT

This study developed an intermittent oxidation-reduction potential (ORP)-controlled micro-aeration system for high solids anaerobic digestion (AD) of lignocellulosic biomass without volatile fatty acids (VFA) accumulation at high organic loading rate (OLR). Traditional AD of Napier grass, a model lignocellulosic biomass, at an OLR of 5 g volatile solids (VS)/L/day resulted in an accumulation of total VFA concentration up to 9.2 g/L as acetic acid (HAc) equivalent, causing rapid drops in pH and methane yield, and driving the digester to the verge of failure. Once intermittent (every 24 h) ORP-controlled micro-aeration (at ORP of −470 mV) was initiated, the total VFA concentration rapidly decreased to 3.0 g HAc/L and the methane yield improved, resulting in stable digester performance without the need for alkalinity supplementation or OLR reduction. By combining reactor performance results, mass balance analyses, microbial community characterization data, and a bioenergetic evaluation, this study suggested that rapid VFA conversion and CH_4 production were carried out by facultative anaerobes and hydrogenotrophic methanogens under micro-aerobic conditions. This novel operating approach can be applied as an effective control strategy for high OLR AD processes especially in the event of VFA accumulation.
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1. Introduction

Anaerobic digestion (AD) has received increased attention in recent years due to several inherent merits, including its ability to produce renewable energy from diverse organic materials, remediate waste streams, and curtail greenhouse gas emissions (Surendra et al., 2015). Limitations of the AD process, however, are rooted in the intrinsic difficulty of maintaining a balance among the activities of hydrolytic, acidogenic, acetogenic, and methanogenic microorganisms. Slow digestibility, especially when fed with recalcitrant substrates such as lignocellulosic biomass, is the main reason for low methane yield (Sawatdeenarunat et al., 2015). To overcome this limitation, digesters are commonly operated at high organic loading rates (OLRs) using feedstocks with a high total solids (TS) content. However, at high OLRs, digesters are susceptible to failure due to the accumulation of volatile fatty acids (VFA), mainly acetic, propionic, and butyric acids, caused by an imbalance between syntrophic acetogenic bacteria and methanogens (Boe and Angelidaki, 2012; Ward et al., 2008). This syntrophic relationship is extremely sensitive since the hydrogen partial pressure (pH_2) has to be low enough for the anaerobic degradation of VFA to be thermodynamically favorable, but high enough for the consumption by H_2-utilizing methanogens (Dolfing, 2014; Labib et al., 1993). Failure to maintain this delicate balance leads to AD process failure due to VFA build-up, and may require restarting of the digester thereby causing significant economic loss (Dong et al., 2011; Steyer et al., 2006). Hence, there is a critical need to develop effective control strategies to maintain stability and rapid recovery of AD processes from potential failure with simultaneous enhancement of hydrolysis, VFA utilization, and methane production. To this end, we developed a unique approach of improving digester stability by intermittently injecting small amount of oxygen (micro-aeration), the timing of which is controlled by oxidation-reduction potential (ORP).

Traditionally, oxygen or air exposure is avoided in AD systems to minimize its negative impact on anaerobic microorganisms.

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https://doi.org/10.1016/j.watres.2019.115080
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However, methanogenesis was recently found to occur outside of anaerobic environments as well, such as in oxygenated freshwater and soil (Angle et al., 2017). Recent studies have also shown that micro-aeration in anaerobic processes can enhance hydrolysis rate, methane yield, VFA production, and hydrogen sulfide removal (Nguyen and Khanal, 2018). The underlying rationale for the beneficial effects of micro-aeration on AD processes is the augmentation in diversity and activity of facultative bacteria that promote hydrolysis, fermentation, and sulfide oxidation (Krayzelova et al., 2015; Lim et al., 2013; Zhu et al., 2009). Based on the above reasoning, we hypothesized that VFA accumulated in the digesters during overload conditions would be rapidly consumed by facultative heterotrophs under micro-aerobic conditions, thus contributing to AD process stability.

To use micro-aeration as an effective operational strategy in AD, precise oxygen dosing control is needed to prevent overdosing and inhibition of anaerobic metabolisms. Very limited studies have focused on an effective micro-aeration dosing system that can be applied in the AD processes (Nguyen and Khanal, 2018). Oxidation-reduction potential (ORP) or redox potential, which exhibits a logarithmic relationship with dissolved oxygen concentration (Fig. S1), can be used for precise oxygen dosing control during micro-aeration. This approach was previously applied in biological hydrogen sulfide removal (Khanal and Huang, 2003a,b) and in pure culture fermentation processes to augment the product yield (Liu et al., 2013). However, insights into the effects of ORP-controlled micro-aeration on metabolic pathways and synergistic interactions between aerobic-anaerobic and bacterial-archaeal populations in AD process are still lacking.

The overall goal of this study was to develop and validate the performance of an ORP-based micro-aeration system to prevent VFA accumulation without the inhibition of methanogenesis during AD of lignocellulosic feedstock at high OLR. After investigating the effects of increasing OLR on AD performance, ORP-based micro-aeration was initiated to examine for AD process stability control. By combining reactor performance and microbial community analyses with bioenergetics and mass balance calculations, we provide new insights into the dynamics of ORP-controlled micro-aeration in AD process and present an effective strategy for maintaining the stability of a highly loaded AD system.

2. Materials and methods

2.1. Feedstock and inoculum

Napier grass (*Pennisetum purpureum*, 5 months old), a typical energy crop, was chosen as a representative lignocellulosic feedstock in this study. The biomass was harvested from the Waimanalo Research Station (Waimanalo, HI, USA). The freshly harvested biomass was passed through a shredder (Vincent Corporation, Tampa, FL, USA) and was air-dried until the moisture content reached <15% for stable storage at room temperature. The dried biomass was then passed through a cutting mill (Retch SM 2000, Haan, Germany) with a screen size of 2 mm. The milled biomass was stored in vacuum bags at room temperature and was used as the sole feedstock for the entire research period.

Cattle manure-derived inoculum, maintained in a 20 L inoculum reactor at mesophilic condition, was used to start up an anaerobic reactor. The inoculum reactor had been fed with fresh cattle manure for 40 days and the digestate was passed through a #8 sieve (ASTM 2.36 mm, Thermo Fisher Scientific Inc., USA) to remove residual fibers. This inoculum was then stored anaerobically at 4 °C until further use. The processed feedstock and inoculum were characterized for TS, volatile solids (VS), fiber content (hemicellulose, cellulose, and lignin), carbon-to-nitrogen (C/N) ratio, and total VFA-to-total alkalinity (VFA/ALK) ratio (Table S1).

2.2. Bioreactor set-up and operation

A continuous-stirred tank reactor (CSTR) with a working volume of 2 L was fabricated using acrylic and equipped with an ORP-based micro-aeration control system and a rubber heating jacket with temperature controller (Power density: 1.25 W/inch^2, Brisk Heat, OH, USA) to maintain the reactor temperature at 35 ± 1 °C (Fig. 1). The experiment was performed three times (Replicates 1, 2, and 3) with the same inoculum, substrate, and reactor conditions to confirm the performance of the ORP-based micro-aeration system, and the results (mean value) are summarized in Table 1. The performance of the reactor in Replicate 1 is primarily discussed in the main manuscript and the results from Replicates 2 and 3 are given in Appendix A.

The bioreactor in Replicate 1 was operated anaerobically at OLRs of 1.5, 3.0, and 5.0 g VS/L/day from days 1–30, 42–60, and 75–116, respectively (Table S6). Between each of these three operating periods, the OLR was increased in a step-wise increment of 0.5 g VS/L/day, till stable performance (a quasi-steady state) was achieved (no significant difference (ANOVA, $p > 0.05$) in each monitoring parameter during five consecutive days of operation). In this study, the term "stable" (with above definition) is used instead of "quasi-steady state" for brevity. Alkalinity was periodically supplemented by adding $NaHCO_3$ (mainly under anaerobic conditions at high OLR) to maintain the pH above 6.0. The reactor was operated in a semi-continuous mode by daily withdrawal of digestate followed by feeding a mixture of biomass (Napier grass) and distilled water to maintain desired OLRs and constant hydraulic retention time (HRT) of 20 days. ORP-based micro-aeration was initiated on day 117 in Replicate 1 at OLR 5.0 g VS/L/day when the bioreactor showed signs of instability, and the performance of this bioreactor was further divided into four periods for comparison, namely: 5.0 stable (days 75–94: anaerobic condition before rapid VFA accumulation), "5.0 unstable (days 95–116: anaerobic condition with rapid VFA accumulation, pH drop, and low methane yield regardless of alkalinity additions), 5.0 micro-aerobic (days 117–155: intermittent micro-aeration condition), 5.0 anaerobic after micro-aeration (days 156–242: anaerobic condition after stopping the micro-aeration system).

2.3. ORP-based micro-aeration system

ORP, which is highly sensitive to oxygen (Fig. S1), was selected as the parameter to control micro-aeration. The bioreactor was equipped with a thermocouple (TTSS-116U-12, Omega Engineering, CT, USA) and an ORP probe (EW-27003-40, Cole-Parmer, IL, USA) to monitor temperature and ORP, respectively, using data loggers (Dataq, OH, USA). The ORP probe was connected to an ORP controller (EW-56700-00, Cole-Parmer, IL, USA) to control the micro-aeration system. The flow rate of oxygen from the compressor tank (OX R200, Airgas, HI, USA) was finely adjusted at 1 mL/min using a two-staged low-pressure regulator and a needle valve. A digital timer was used to automatically monitor the total micro-aeration time in seconds. The reactor was subjected to micro-aeration every other day (Fig. S2). The micro-aeration intensity (L O_2/L reactor/day) was calculated by multiplying the oxygen flow rate with the daily micro-aeration time and divided by the reactor working volume. Details of the ORP-controlled micro-aeration system are provided in Appendix A.

2.4. Chemical analyses

pH, ORP, and methane yield were monitored daily; individual VFA and VFA/ALK ratio were analysed every two days; TS, VS, and

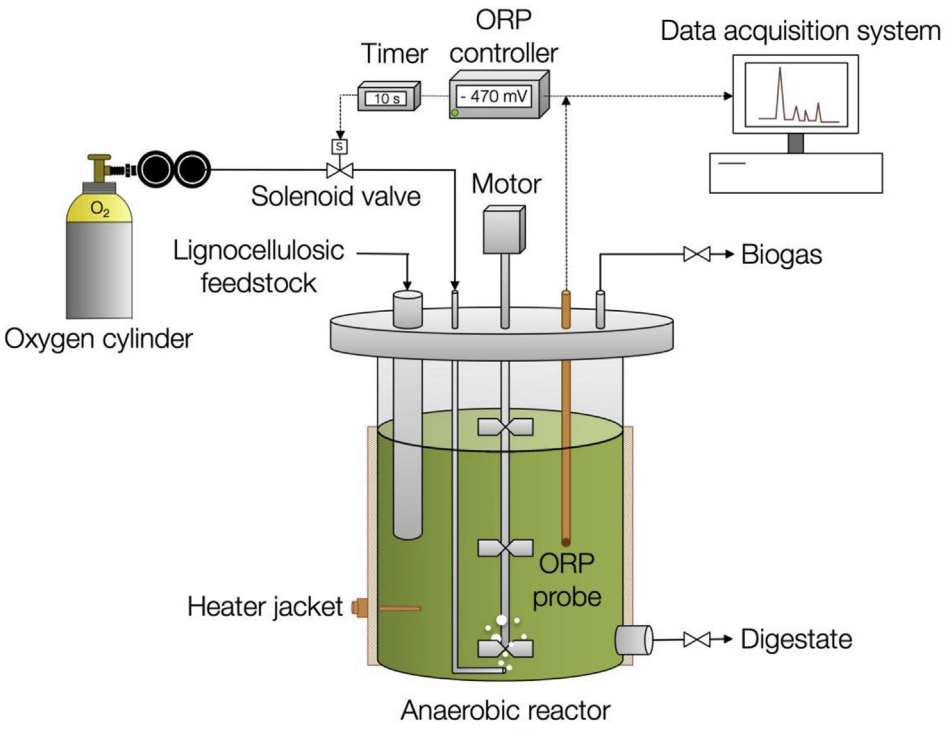


Fig. 1. Schematic diagram of bioreactor set-up with ORP-based micro-aeration system.

fiber compositions were analysed weekly. Details of the chemical analyses are provided in Appendix A.

2.5. Microbial analyses

Biomass samples under anaerobic and micro-aerobic conditions were collected from the bioreactor in Replicate 1 on days 79 and 149, Replicate 2 on days 150 and 192, and Replicate 3 on days 64 and 108, respectively, when the OLR was 5 g VS/L/day. The collected biomass samples were subjected to DNA extraction using the Power Soil DNA Isolation Kit (MO BIO Laboratories, Inc., Canada) as per the manufacturer's instructions.

The bacterial and methanogenic communities were investigated by targeting the 16S rRNA gene using 515F/806R primers and the *mcrA* gene using mcrA3F/mcrA3R primers, respectively (Caporaso et al., 2012; Luton et al., 2002). Sequencing was performed using the Illumina HiSeq2000 platform and sequence reads were processed using mothur (version 1.35.0) (Schloss et al., 2009). Chimeras were removed using UCHIME (Edgar et al., 2011). Bacterial 16S rRNA gene fragments were aligned against the SILVA database (release 123) (Pruesse et al., 2007) and *mcrA* gene fragments against a subset of the Fun Gene database (Fish et al., 2013), resulting in taxonomic labels for each aligned fragment. The raw sequence reads were deposited in the NCBI Sequence Read Archive (SRA; http://www.ncbi.nlm.nih.gov/sra) under accession numbers SAMN12752316 to SAMN12752321 and SAMN12752323 to SAMN12752328.

2.6. Statistical analyses

Statistical significance was determined by analysis of variance (ANOVA). Tukey's Honest Significant Difference (HSD) test was used to compare differences among various bioreactor operating conditions (Table 1). All statistical analyses were carried out using JMP software (JMP Pro Version 12.0, SAS Institute Inc., NC, USA) with p-value of 0.05.

3. Results

3.1. Bioreactor performance without micro-aeration at OLR of 1.5, 3.0 and 5.0 g VS/L/day

Increasing the OLR resulted in significant increases in total VFA, individual VFAs (acetate, propionate, and butyrate), VFA/ALK ratio, and TS content in the reactor with concomitant decreases in methane yield, pH and VS removal efficiency. Stable operation was achieved at OLRs of 1.5 and 3.0 g VS/L/day with an average methane yield of 102–105 ± 4 NmL/g VS added, which was comparable to other studies using the same substrate (Sawatdeenarunat et al., 2015; Surendra and Khanal, 2014). When the OLR was increased to 5.0 g VS/L/day, the reactor performance began to deteriorate (Table 1).

In Replicate 1, the bioreactor operating at an OLR of 5.0 g VS/L/day from day 75, showed a rapid accumulation of VFAs (Fig. 2B and C), which led to decreases in pH and methane yield (Fig. 2A and B). The bioreactor was particularly unstable from days 95–116 as the total VFA had increased to >9.0 g acetic acid equivalent (HAc)/L, the VFA/ALK ratio was >2.0, and the pH had dropped to < 5.5, resulting in a reduction in the methane yield to as low as 5.0 NmL/g VS regardless of attempts to stabilize the bioreactor through alkalinity supplementations (Fig. 2A). The results demonstrated that mono-digestion of Napier grass under mesophilic conditions started to experience organic overloading at OLR 5.0 g VS/L/day. To overcome this limitation, the effectiveness of intermittent ORP-controlled micro-aeration (24 h every other day) to obtain stable bioreactor performance under high OLR was examined.

3.2. Bioreactor performance with intermittent ORP-controlled micro-aeration at OLR of 5.0 g VS/L/day

The intermittent ORP-controlled micro-aeration operation was started on day 117 in Replicate 1 (Fig. 2). Within 12 days, the total

Table 1
Reactor performance at various organic loading rates and conditions for Replicates 1, 2 and 3.

	Organic loading rate (g VS/L/day)*				
	1.5	3.0	5.0	5.0 unstable	5.0 micro-aerobic
pH	6.97 ± 0.08a	6.86 ± 0.08ab	6.73 ± 0.08b	6.43± 0.08c	6.98 ± 0.08a
ORP (mV-E_{Ag})	-473 ± 4a	-500 ± 4b	-503 ± 4b	-465 ± 4a	-464 ± 6a
Methane yield (NmL/g VS added)	104.5 ± 3.7a	101.9± 3.7a	80.9 ± 3.7b	22.9 ± 3.7c	78.1 ± 3.7b
CH_4/CO_2 ratio (v/v)	1.27 ± 0.03a	1.22 ± 0.03a	1.03 ± 0.03b	0.72 ± 0.03c	1.08 ± 0.03b
Total VFA (g/L as HAc)	0.32 ± 0.40c	1.13 ± 0.32c	3.01 ± 0.35b	7.37 ± 0.31a	3.50 ± 0.31b
Total alkalinity (g/L as $CaCO_3$)	2.02 ± 0.38c	2.04 ± 0.31c	2.88 ± 0.34bc	3.88 ± 0.31a	3.59 ± 0.31ab
VFA/ALK ratio (g HAc equivalent to g $CaCO_3$ equivalent)	0.15 ± 0.2c	0.55 ± 0.15bc	1.05 ± 0.17b	2.16 ± 0.14a	0.95 ± 0.15b
Acetic acid (g/L)	0.15 ± 0.22c	0.21 ± 0.17c	0.52 ± 0.6bc	2.77 ± 0.16a	0.70 ± 0.17b
Propionic acid (g/L)	0.15 ± 0.25c	0.66 ± 0.18c	1.75 ± 0.17b	2.43 ± 0.17a	1.81 ± 0.18b
Butyric acid (g/L)	0.00 ± 0.06b	0.13 ± 0.05b	0.17 ± 0.05b	0.93 ± 0.05a	0.13 ± 0.06b
Digestate TS (%)	2.6 ± 0.5d	5.4 ± 0.5c	7.1 ± 0.4b	10.8 ± 0.4a	8.0 ± 0.4b
VS removal (%)	60.7 ± 3.0a	51.3 ± 3.0ab	35.2 ± 2.9c	20.6 ± 2.7d	47.3 ± 2.7b
Digestate hemicellulose (% of TS)	22.77 ± 0.47a	24.33 ± 0.47a	23.19 ± 0.47a	22.96 ± 0.47a	22.41 ± 0.47a
Digestate cellulose (% of TS)	51.14 ± 0.85a	53.26 ± 0.85a	55.44 ± 0.85a	55.15 ± 0.85a	54.22 ± 0.85a

*Data shown as mean ± standard error of triplicate experiments with five replicate measurements for each parameter (n = 15). Data in each row with different letters ("a"-highest value and "d"-lowest value) indicate significant differences (Tukey's HSD test with $p < 0.05$), the darker shades (significant difference from Tukey's test) represent undesirable values and indicate instabilities.

VFA concentration decreased from 9.2 to 3.0 g HAc/L and the acetic acid concentration dropped from 4.0 g/L to less than 0.7 g/L (Fig. 2C) with a concomitant increase in methane yield from 40 to 114 NmL/g VS (Fig. 2B). All other important parameters stabilized as well (Fig. 2), suggesting full recovery of the bioreactor. Stable operation of the bioreactor at high OLR continued even under anaerobic conditions (from days 156–215) after intermittent micro-aeration was terminated. However, from days 215–242, the gradual accumulation of VFA was observed, likely leading to another unstable period under anaerobic condition.

Replicates 2 and 3 confirmed the effectiveness of ORP-controlled micro-aeration on digester stability. The recovery trends in Replicates 2 and 3 (Figs. S3 and S4) were similar to those in Replicate 1. For example, a drop in the total VFA level from 10.8 to 5.0 g HAc/L and from 9.2 to 2.1 g HAc/L with concomitant increase in methane yields from 5 to 102 NmL/g VS and from 21 to 125 NmL/g VS were observed after micro-aeration was initiated in Replicates 2 and 3, respectively (Figs. S3 and S4). These replicate experiments confirmed the effectiveness of intermittent micro-aeration to control VFA accumulation and to maintain bioreactor stability. The average values of important performance parameters for the three replicates are summarized in Table 1.

Intermittent micro-aeration was achieved by controlling the ORP at a setpoint of −470 mV (+25 mV above the anaerobic baseline ORP of −495 mV, Fig. 2E) for a 24-h period, followed by a 24-h period without micro-aeration. The ORP levels varied from an average of −468 ± 4 mV to an average of −522 ± 12 mV, during the days with and without micro-aeration, respectively, indicating a rapid exhaustion of residual dissolved oxygen (DO) and a switching back to anaerobic conditions. Also, the average daily ORP of the anaerobic period after micro-aeration (days 156–242) remained fairly stable at a baseline ORP level of −530 ± 9 mV (Fig. 2E). The ORP levels during anaerobic and micro-aerobic conditions were in close agreement with those in previous studies (Jenicek et al., 2011;

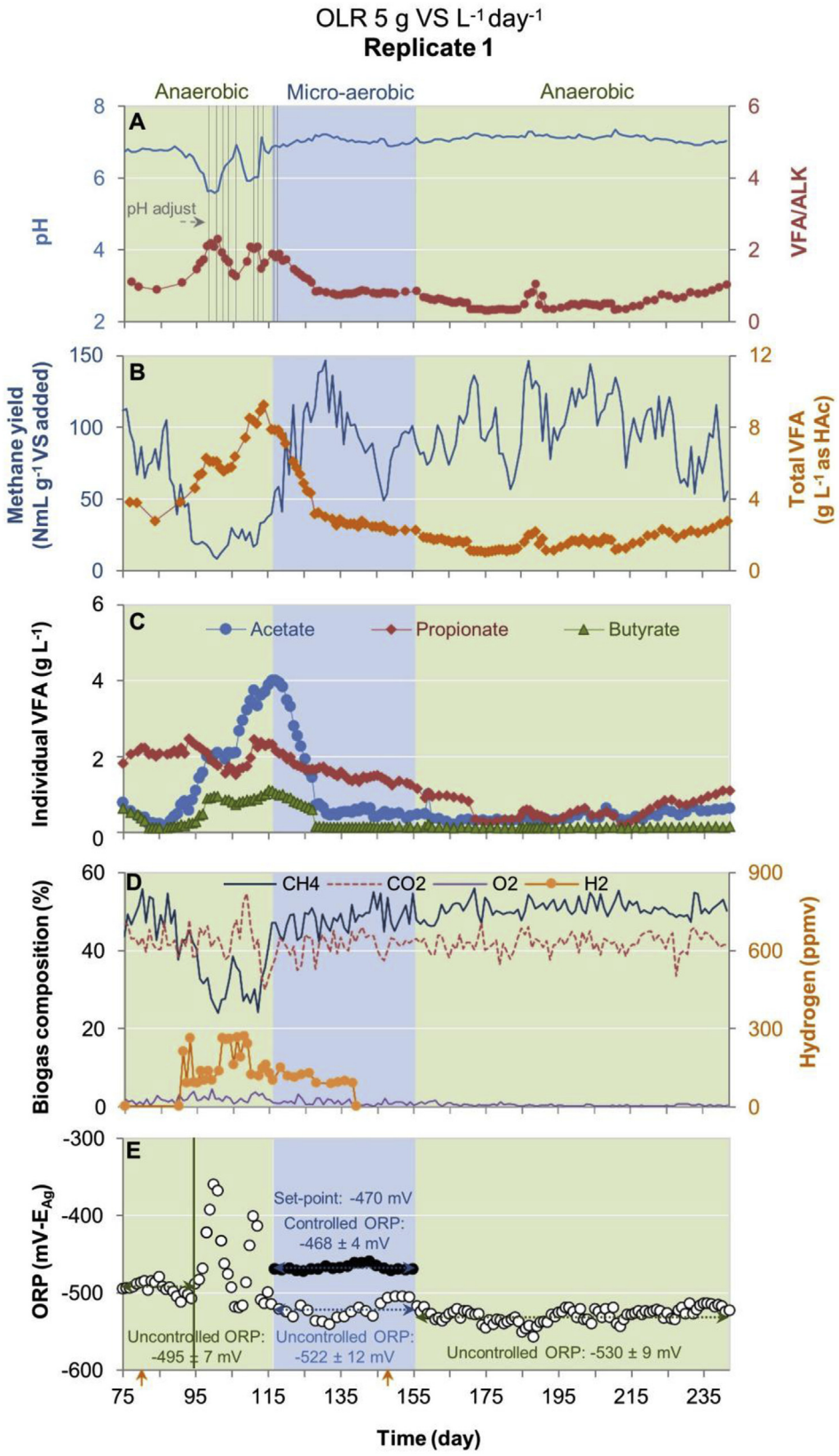


Fig. 2. Reactor instability and recovery by micro-aeration at an OLR of 5.0 g VS/L/day for Replicate 1 (days 75–242). Process performance in terms of pH and VFA/ALK ratio with vertical lines indicating additions of $NaHCO_3$ to control pH (A); total VFA and methane yield (B); individual VFA concentrations (iso-butyric, valeric, and iso-valeric acid concentrations were below 0.2 g/L at all times and are not shown here) (C); biogas composition (D); and daily average ORP of uncontrolled ORP (open circles) and controlled ORP every other day (closed circles) (E). The mean (±standard deviation) uncontrolled ORP from days 75–94 (anaerobic stable period, before pH fluctuation) was calculated to be −495 ± 7 mV. ORP from days 95–116 was disregarded due to fluctuations resulting from pH control during this period. The mean controlled ORP (−468 ± 4 mV) was the average of the ORP values of days with ORP-based micro-aeration (closed circles) from days 117–155. Micro-aeration was performed every other day, thus the mean of uncontrolled ORP values (open circles) from days 117–155 was −522 ± 12 mV, representing the daily average ORP on the days without micro-aeration. The mean value of uncontrolled ORP (anaerobic condition) after the micro-aeration period (days 156–242) was −530 ± 9 mV. Orange arrows in the x-axis indicate the days that biomass samples were taken for microbial community analysis (days 79 and 149). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Khanal and Huang, 2006). Thus, intermittent ORP-based micro-aeration was instrumental in digester recovery from the verge of failure without supplementation of alkalinity and/or reducing the OLR.

3.3. Micro-aeration enhanced carbon recovery

The results of a carbon mass balance performed before and after micro-aeration in three replicate experiments (Fig. 3) indicate that micro-aerobic conditions enhanced lignocellulosic degradation and carbon recovery in the form of biogas (CO_2 and CH_4) production. Specifically, the digestate from the micro-aerated condition consisted of 2.84 ± 0.13 g carbon, which was significantly lower ($p < 0.05$) than 3.66 ± 0.13 g carbon present in the digestate without micro-aeration. This finding was further supported by a significant increase ($p < 0.05$) in VS reduction from $20.6 \pm 2.7\%$ during the anaerobic unstable period to $47.3 \pm 2.7\%$ after micro-aeration was initiated (Table 1). This increase in substrate degradation resulted in higher soluble substrate concentrations for microbial communities and therefore significantly improved biogas production and enhanced carbon recovery in the form of CO_2 and CH_4. In addition, the removal of accumulated VFA stabilized the pH of the bioreactor and facilitated the quick recovery of the activity of methanogens thereby resulting in significant enhancement in CH_4 production ($p < 0.05$). However, the carbon mass balance in this study could not identify the fractions of carbon that went into each pathway within the AD process (aerobic respiration or anaerobic syntrophic acetogenesis; acetoclastic or hydrogenotrophic methanogenesis). Such information could only be obtained from carbon isotope labelling techniques (Gehring et al., 2015), which was beyond the scope of this study.

3.4. Micro-aeration promoted VFA conversion by facultative bacteria through aerobic respiration

In traditional AD process, the syntrophic interaction between acetogens and methanogens is critically important to maintain low VFA concentrations in the aqueous phase (Leng et al., 2017a; Yuan et al., 2011). However, due to thermodynamic limitation of syntrophic acetogenic reactions (Leng et al., 2017b), this rate-limiting step in AD process often causes VFA build-up and leads to reactor failure as is the case in this study.

This study suggested a dominant mechanism of effective VFA conversion by facultative bacteria and methane production by hydrogenotrophic methanogens under controlled micro-aerobic conditions. The suggestion is based on mass balance analyses, thermodynamic calculation, and the three observations drawn from the microbial community analysis: (1) the relative abundance of syntrophic bacteria remained low (<1.0% of total bacteria), (2) the relative abundance of facultative bacteria increased after ORP-based micro-aeration, and (3) methanogenesis was achieved mainly by hydrogenotrophic methanogens. These observations are discussed in greater detail in the following paragraph.

The high OLR of 5.0 g VS/L/day resulted in VFA accumulation, and drops in pH and methane production (Table 1), suggesting a microbial imbalance before micro-aeration was initiated. The microbial analyses confirmed that syntrophic bacteria (family *Syntrophaceae and Syntrophomonadaceae*) were present at low relative abundances under anaerobic condition as their combined relative abundance accounted for less than 1.0% of total bacteria in all three replicate experiments (Table S2). Similar microbial results were reported for AD systems subjected to stress or high VFA concentrations (Feng et al., 2009; Goux et al., 2015; Luo et al., 2016; Treu et al., 2016). The relative abundance of these syntrophic acetogens remained below 1.0% (Table S2) after implementing micro-aeration. At the same time, there was a significant increase in relative abundance of facultative bacteria (Fig. 4), especially the most abundant operational taxonomy units (OTUs) associated with the genus *Proteiniphilum* (Chen and Dong, 2005) increased from 16% to 32% following micro-aeration. The increase in these OTUs was even more distinct in Replicates 2 (from 2% to 65%) and 3 (from 1% to 43%) (Table S2), indicating that *Proteiniphilum* spp. played an important role in converting excess VFA into CO_2 during intermittent ORP-based micro-aeration in AD process. Additionally, the dominance of hydrogenotrophic methanogens was observed in all samples obtained during micro-aeration for all three replicate experiments, suggesting that the increase in methane production was associated with CO_2 generated during the preceding aerobic respiration reactions.

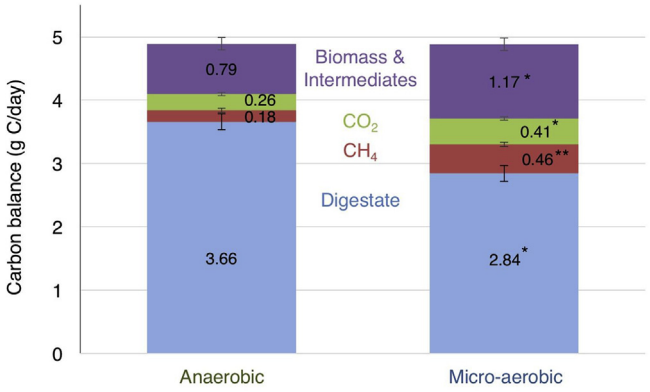


Fig. 3. Carbon mass balance under anaerobic and micro-aerobic conditions.
Note: Data are represented as mean values (with standard errors) of triplicate samples from three replicate experiments (n = 9 for each condition). Significant differences from ANOVA test with $p < 0.05$ (*) and $p < 0.005$ (**). The mass balance calculation method is provided in Appendix A.

To further reinforce the findings suggested by the microbial community analyses, thermodynamic calculations were performed (Fig. 5). Thermodynamic results show that the changes in Gibbs free energy associated with complete VFA oxidation to CO_2 using O_2 as an electron acceptor ($\Delta G'_R = -91.7$ to -88.2 kJ/e$^-$) were much higher than that of incomplete VFA fermentation coupled with proton reduction ($\Delta G'_R = -11.1$ to $+6.9$ kJ/e$^-$) at the actual conditions of Replicate 1 (Fig. 5). These thermodynamic values at the actual conditions are also more conservative compared to values at standard conditions where VFA aerobic oxidation reactions are even more thermodynamically favorable ($\Delta G^{o'} = -109$ kJ/e$^-$) than those for the VFA anaerobic fermentation reactions ($\Delta G^{o'} = -1$ to $+23$ kJ/e$^-$) (Table S3).

These results suggest that the facultative bacteria switched from anaerobic fermentation to the more thermodynamically and kinetically favorable aerobic respiration (Tables S3 and S4). Upon the dosing of O_2 for the facultative bacteria, the more energetically favorable aerobic respiration could be regulated via specific gene expression (Forster and Gescher, 2014; Morris and Schmidt, 2013). Thus, the VFA produced during the period without micro-aeration were likely consumed by the facultative heterotrophs when micro-aeration was initiated, resulting in low VFA concentrations during the period with intermittent micro-aeration (Fig. 2). This increases in relative abundance of facultative bacteria while preserving methanogens for a more balanced digestion with a long-lasting effect even after the micro-aeration was terminated. It, therefore, triggered the stable operation of the reactor under anaerobic condition from days 156–242 (Fig. 2). Moreover, after the build-up of VFA was reduced to the baseline level (around 3 g/L as HAc) via ORP-based micro-aeration, the reactor could be operated

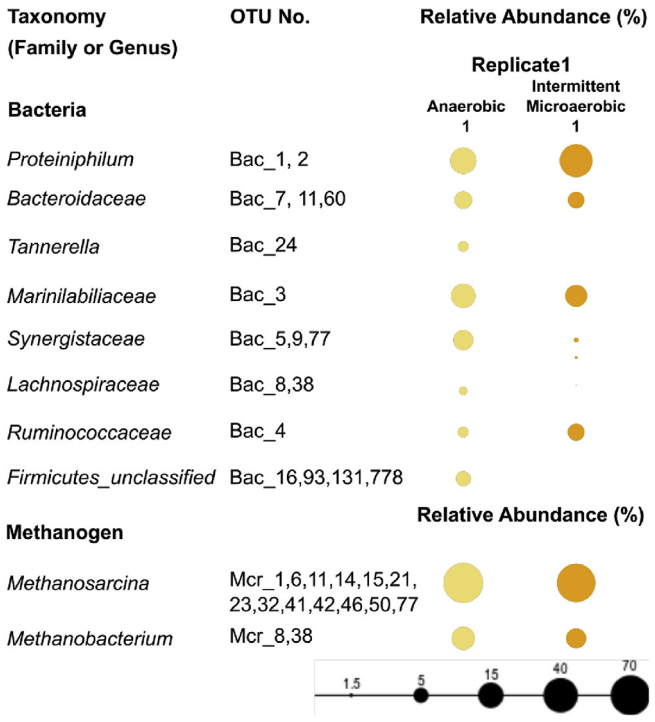


Fig. 4. Bacterial community structure (A) and archaeal community structure (B) in Replicate 1 without (day 79) and with (day 149) intermittent ORP-controlled micro-aeration. Only OTUs with relative abundance >2% at genus level are included. Details of relative abundance and potential function of identified OTUs are presented in Table S2. Bacterial and community structure of Replicate 2 and 3 are presented in Fig. S5.

at anaerobic condition without micro-aeration control for a long period through the newly established microbial community (Fig. S6). Nonetheless, the duration in which the reactor could be operated anaerobically without micro-aeration varied (e.g., the stable period after stopping micro-aeration in Replicate 3 was relatively short (Fig. S4)), possibly due to ecologically dynamic responses (Axelrod and Hamilton, 1981).

Hence, this intermittent micro-aeration system had two purposes: the first was to provide the precise amount of O_2 for facultative heterotrophs to rapidly consume the accumulated VFA during unstable periods, and the second was to develop a new balanced microbial community for long-term stable methane production at high OLRs. While alkalinity addition for pH control only caused transient relief of VFA inhibition to methanogens, the intermittent micro-aeration system effectively scavenged excess VFA, thereby maintaining digester stability during high organic loading conditions.

3.5. ORP-controlled micro-aeration maintained anaerobic niches for methanogenesis

Although intermittent ORP-based micro-aeration created micro-aerobic niches for VFA conversion by facultative microbes, it preserved the crucial methanogens as apparent from the increased methane production (Figs. 2 and 3), indicating a vibrant methanogenic activity. Even though the accumulated VFA were likely oxidized to CO_2 through aerobic respiration during micro-aeration, there was no significant change ($p > 0.05$) in the CO_2 content of the biogas, suggesting that CO_2 was reduced to CH_4 using H_2 as an electron donor. This was further supported by the finding that the most abundant methanogenic populations under intermittent micro-aeration conditions were all hydrogenotrophic methanogens (Table S2).

In Replicate 1, due to the abundance of the metabolically versatile genus *Methanosarcina* (CH_4 production via H_2 or acetate (De Vrieze et al., 2012)), a flexible metabolic switch from acetoclastic to hydrogenotrophic methanogenesis may have occurred during micro-aeration without population succession (Fig. 4). In Replicates 2 and 3, the strictly hydrogenotrophic methanogens belonging to the genera *Methanobrevibacter* (Leadbetter and Breznak, 1996) and *Methanocorpusculum* (Gilmore et al., 2017) dominated under micro-aerobic conditions, confirming that ORP-based micro-aeration favoured hydrogenotrophic methanogenesis over acetoclastic methanogenesis (Table S2). Methanogenic genera *Methanosarcina* and *Methanobrevibacter* are known to possess oxygen detoxifying enzymes, such as superoxide dismutase (SOD), catalase and peroxidase, which might be responsible for oxygen tolerance in micro-aeration conditions (Nguyen and Khanal, 2018).

Moreover, although H_2 oxidation with O_2 to H_2O ($\Delta G'_R = -92.7$ kJ/e$^-$) is thermodynamically more favorable than H_2 reduction to CH_4 ($\Delta G'_R = -4.0$ kJ/e$^-$), the supplied O_2 was likely completely consumed during VFA oxidation, thereby leaving the available H_2 for hydrogenotrophic methanogens. Here, the fast-growing facultative heterotrophs ($\mu_m = 13.2$ 1/day) appeared to have an advantage in the presence of excess VFA over slow-growing H_2-oxidizing bacteria ($\mu_m = 10.1$ 1/day) with limited H_2 availability, and with similar Gibbs free energy changes for aerobic oxidation of VFA and H_2 ($\Delta G'_R = -88.2$ to -92.7 kJ/e$^-$) (Tables S3 and S4). Consistent with this, the OTUs associated with facultative heterotrophs accounted for more than 30% of the total bacteria, while H_2-oxidizing bacteria were not detected among the dominant OTUs (Table S2).

As shown in Fig. 2, the reduction of H_2 to an undetectable level in the biogas occurred after day 140, when the total VFA concentration was as low as 3.0 g HAc/L. This could be due to aerobic oxidation of H_2 to H_2O, whereas prior to day 140, all supplied O_2 was likely consumed by facultative heterotrophs to oxidize accumulated VFA to CO_2. Our results suggest that intermittent micro-aeration by controlling ORP level at -470 mV was insufficient to fully support aerobic growth; thus facilitating anaerobic metabolism for methanogens in such micro-aerobic environment.

Further, in an independent set of experiments, it was observed that continuous micro-aeration at an ORP +25 mV higher than the baseline ORP and intermittent micro-aeration at an ORP +50 mV higher than the baseline ORP caused significant accumulation of VFA due to inhibition of methanogens as indicated by reduced methane yields (Table S5). The higher oxidative micro-environment was unfavourable to support the co-existence of facultative bacteria and anaerobic methanogens (Liu et al., 2013). Other studies have also reported that micro-aeration at an ORP greater than +25 mV (e.g., +50 or +100 mV) above the baseline level caused adverse effects on reactor performance likely due to excess aerobic respiration by facultative heterotrophs (Khanal and Huang, 2003a,b, 2006).

4. Discussion

Our results suggest that ORP-controlled intermittent micro-aeration promoted VFA conversion by facultative bacteria while conserving crucial anaerobic niches for methanogenesis, allowing stable operation of AD reactors at high OLR. Combining reactor performance results, mass balance analyses, microbial community characterization data, and bioenergetics evaluations, we propose a putative methane producing pathway for complex substrates such as lignocellulosic biomass at high OLR under intermittent ORP-controlled micro-aerobic conditions as illustrated in Fig. 6. However, further metatranscriptomic analyses are needed to evaluate

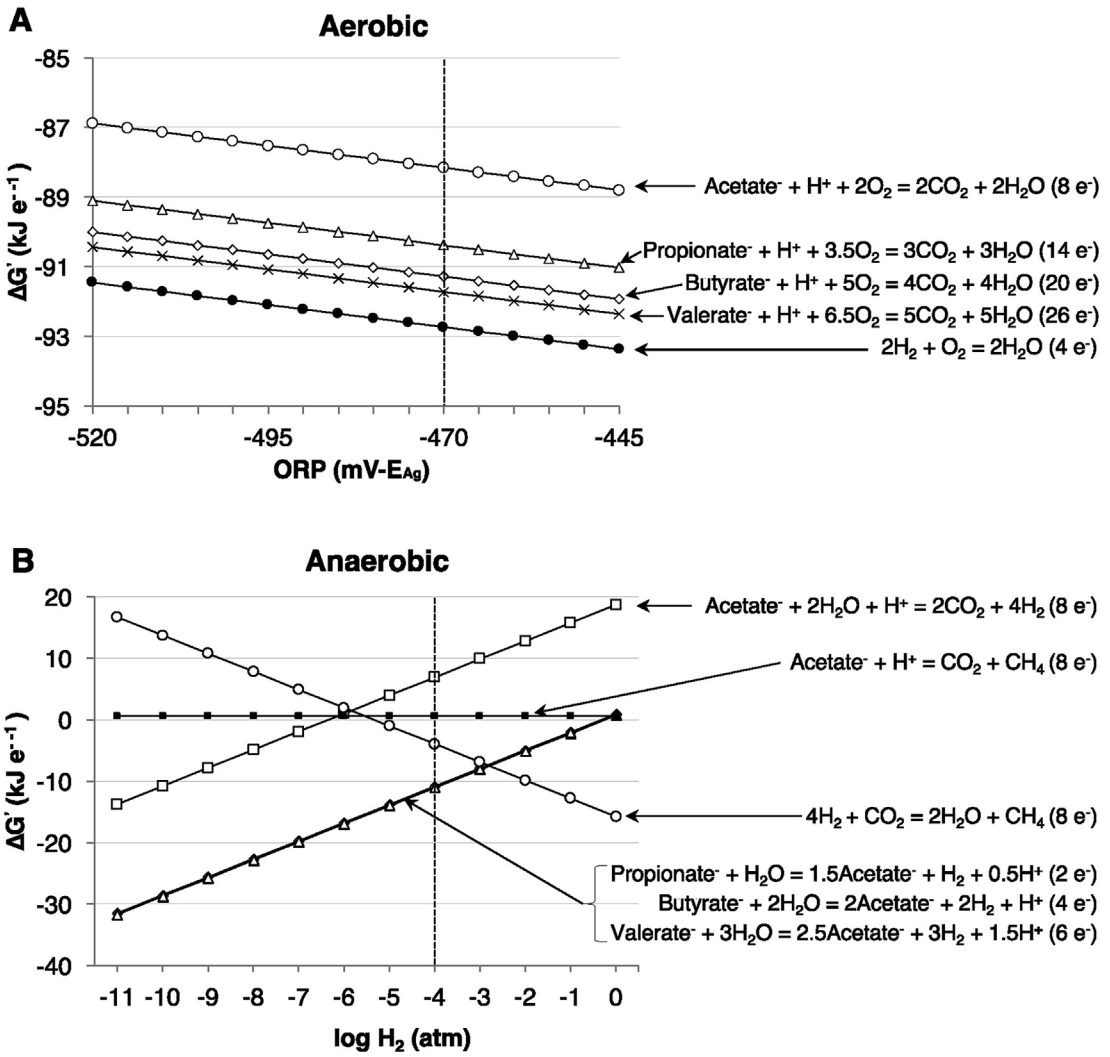


Fig. 5. Gibbs free energy changes at actual condition (ΔG′) in Replicate 1 of aerobic reactions as a function of ORP (A) and anaerobic reactions as a function of log H_2 partial pressures (B). Legends are stoichiometric equations with the number of electrons transferred in parentheses. Calculations were made according to (Thauer et al., 1977) with all standard values of the aqueous phase. Details of calculation method and values are available in Appendix A and Table S3. Vertical lines indicate the bioreactor ORP and H_2 partial pressure under ORP-controlled micro-aerobic conditions.

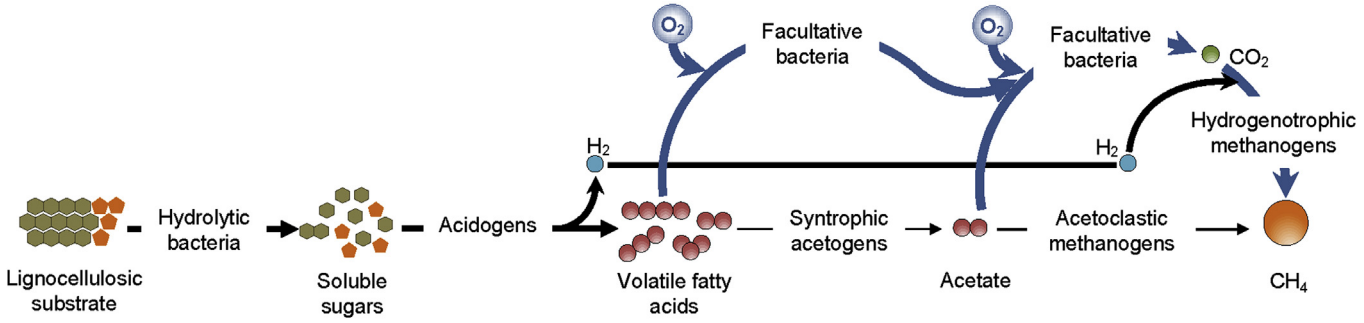


Fig. 6. Proposed methane producing pathway from lignocellulosic biomass at high OLR via intermittent ORP-based micro-aeration.

the specific transcriptional activities in this complex microbial community to confirm the proposed pathway. This novel operational strategy via micro-aeration can be further developed and applied to operate anaerobic digesters at high OLRs. This approach avoids chemical addition for pH control and has significant merits for full-scale implementation of anaerobic digestion of highly complex substrates such as lignocellulosic-based feedstocks.

5. Conclusions

ORP-based micro-aeration was developed and tested for AD of highly complex feedstocks such as lignocellulosic biomass at high

OLR. By increasing the OLR to 5.0 g VS/L/day, a proportional increase in total VFA concentration and a VFA/ALK ratio that caused reduction in pH and methane yield were observed driving the reactor to the verge of failure. When intermittent ORP-controlled micro-aeration was initiated, VFA concentrations rapidly decreased and the reactor performance recovered within two weeks. Controlling the ORP at −470 mV (i.e., +25 mV from the anaerobic baseline ORP) resulted in optimal VFA reduction. The ORP-based micro-aeration provided O_2 as the limiting electron acceptor for facultative heterotrophs allowing partial aerobic oxidation of VFA while conserving crucial anaerobic niches for methanogens. Thus, ORP-based micro-aeration serves as a novel process control tool for highly-loaded AD systems, especially when digester are on the verge of failure due to excess VFA accumulation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This research was funded by the National Institute of Food and Agriculture, U.S. Department of Agriculture (Award number 2013-67022-21177). Wu and Shrestha were supported by funding from Imperial College London, and the University of Michigan and the U.S. National Science Foundation (Sustainability Research Networks 1444745), respectively. Khanal would like to pay special tribute to his Advisor Prof. Ju-Chang (Howard) Huang for introducing ORP application during his graduate studies at the Hong Kong University of Science and Technology (HKUST). Sincere thanks to Ryan Kurasaki for his support in reactor fabrication.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.watres.2019.115080.

References

Angle, J.C., Morin, T.H., Solden, L.M., Narrowe, A.B., Smith, G.J., Borton, M.A., Rey-Sanchez, C., Daly, R.A., Mirfenderesgi, G., Hoyt, D.W., Riley, W.J., Miller, C.S., Bohrer, G., Wrighton, K.C., 2017. Methanogenesis in oxygenated soils is a substantial fraction of wetland methane emissions. Nat. Commun. 8, 1567. https://doi.org/10.1038/s41467-017-01753-4.

Axelrod, R., Hamilton, W.D., 1981. The evolution of cooperation. Science 211, 1390–1396. https://doi.org/10.2307/j.ctvjf9xvk.26.

Boe, K., Angelidaki, I., 2012. Pilot-scale application of an online VFA sensor for monitoring and control of a manure digester. Water Sci. Technol. 66, 2496–2503. https://doi.org/10.2166/wst.2012.498.

Caporaso, J.G., Lauber, C.L., Walters, W. a, Berg-Lyons, D., Huntley, J., Fierer, N., Owens, S.M., Betley, J., Fraser, L., Bauer, M., Gormley, N., Gilbert, J. a, Smith, G., Knight, R., 2012. Ultra-high-throughput microbial community analysis on the Illumina HiSeq and MiSeq platforms. ISME J. 6, 1621–1624. https://doi.org/10.1038/ismej.2012.8.

Chen, S., Dong, X., 2005. *Proteiniphilum acetatigenes* gen. nov., sp. nov., from a UASB reactor treating brewery wastewater. Int. J. Syst. Evol. Microbiol. 55, 2257–2261. https://doi.org/10.1099/ijs.0.63807-0.

De Vrieze, J., Hennebel, T., Boon, N., Verstraete, W., 2012. Methanosarcina: the rediscovered methanogen for heavy duty biomethanation. Bioresour. Technol. 112, 1–9. https://doi.org/10.1016/j.biortech.2012.02.079.

Dolfing, J., 2014. Thermodynamic constraints on syntrophic acetate oxidation. Appl. Environ. Microbiol. 80, 1539–1541. https://doi.org/10.1128/AEM.03312-13.

Dong, F., Zhao, Q.B., Li, W.W., Sheng, G.P., Zhao, J.B., Tang, Y., Yu, H.Q., Kubota, K., Li, Y.Y., Harada, H., 2011. Novel online monitoring and alert system for anaerobic digestion reactors. Environ. Sci. Technol. 45, 9093–9100. https://doi.org/10.1021/es202245f.

Edgar, R.C., Haas, B.J., Clemente, J.C., Quince, C., Knight, R., 2011. UCHIME improves sensitivity and speed of chimera detection. Bioinformatics 27, 2194–2200. https://doi.org/10.1093/bioinformatics/btr381.

Feng, L., Chen, Y., Zheng, X., 2009. Enhancement of waste activated sludge protein conversion and volatile fatty acids accumulation during waste activated sludge anaerobic fermentation by carbohydrate substrate addition: the effect of pH. Environ. Sci. Technol. 43, 4373–4380. https://doi.org/10.1021/es8037142.

Fish, J.A., Chai, B., Wang, Q., Sun, Y., Brown, C.T., Tiedje, J.M., Cole, J.R., 2013. FunGene: the functional gene pipeline and repository. Front. Microbiol. 4, 1–14. https://doi.org/10.3389/fmicb.2013.00291.

Forster, A.H., Gescher, J., 2014. Metabolic engineering of *Escherichia coli* for production of mixed-acid fermentation end products. Front. Bioeng. Biotechnol. 2, 1–12. https://doi.org/10.3389/fbioe.2014.00016.

Gehring, T., Klang, J., Niedermayr, A., Berzio, S., Immenhauser, A., Klocke, M., Wichern, M., Lübken, M., 2015. Determination of methanogenic pathways through carbon isotope (δ13C) analysis for the two-stage anaerobic digestion of high-solids substrates. Environ. Sci. Technol. 49, 4705–4714. https://doi.org/10.1021/es505665z.

Gilmore, S.P., Henske, J.K., Sexton, J.A., Solomon, K.V., Seppälä, S., Yoo, J.I., Huyett, L.M., Pressman, A., Cogan, J.Z., Kivenson, V., Peng, X., Tan, Y.P., Valentine, D.L., O'Malley, M.A., 2017. Genomic analysis of methanogenic archaea reveals a shift towards energy conservation. BMC Genomics 18, 1–14. https://doi.org/10.1186/s12864-017-4036-4.

Goux, X., Calusinska, M., Lemaigre, S., Marynowska, M., Klocke, M., Udelhoven, T., Benizri, E., Delfosse, P., 2015. Microbial community dynamics in replicate anaerobic digesters exposed sequentially to increasing organic loading rate, acidosis, and process recovery. Biotechnol. Biofuels 8, 1–18. https://doi.org/10.1186/s13068-015-0309-9.

Jenicek, P., Celis, C.a., Koubova, J., Pokorna, D., 2011. Comparison of microbial activity in anaerobic and microaerobic digesters. Water Sci. Technol. 63, 2244–2249. https://doi.org/10.2166/wst.2011.579.

Khanal, Huang, J.-C., 2003a. ORP-based oxygenation for sulfide control in anaerobic treatment of high-sulfate wastewater. Water Res. 37, 2053–2062. https://doi.org/10.1016/S0043-1354(02)00618-8.

Khanal, S., Huang, J., 2003b. Anaerobic treatment of high sulfate wastewater with oxygenation to control sulfide toxicity. J. Environ. Eng. 129, 1104–1111.

Khanal, S.K., Huang, J.-C., 2006. Online oxygen control for sulfide oxidation in anaerobic treatment of high-sulfate wastewater. Water Environ. Res. 78, 397–408.

Krayzelova, L., Bartacek, J., Díaz, I., Jeison, D., Volcke, E.I.P., Jenicek, P., 2015. Microaeration for hydrogen sulfide removal during anaerobic treatment: a review. Rev. Environ. Sci. Biotechnol. 14, 703–725. https://doi.org/10.1007/s11157-015-9386-2.

Labib, F., Ferguson, J.F., Benjamin, M.M., Merigh, M., Ricker, N.L., 1993. Mathematical modeling of an anaerobic butyrate degrading consortia: predicting their response to organic overloads. Environ. Sci. Technol. 27, 2673–2684. https://doi.org/10.1021/es00049a005.

Leadbetter, J.R., Breznak, J. a, 1996. Physiological ecology of Methanobrevibacter cuticularis sp . nov . and Methanobrevibacter the termite Reticulitermes flavipes. In: Physiological Ecology of Methanobrevibacter Cuticularis Sp . Nov . And Methanobrevibacter Curvatus Sp . Nov , Isolated from Th 62, pp. 3620–3631.

Leng, L., Yang, P., Mao, Y., Wu, Z., Zhang, T., Lee, P.-H., 2017a. Thermodynamic and physiological study of caproate and 1,3-propanediol co-production through glycerol fermentation and fatty acids chain elongation. Water Res. 114, 200–209. https://doi.org/10.1016/j.watres.2017.02.023.

Leng, L., Yang, P., Singh, S., Zhuang, H., Xu, L., Chen, W.H., Dolfing, J., Li, D., Zhang, Y., Zeng, H., Chu, W., Lee, P.H., 2017b. A review on the bioenergetics of anaerobic microbial metabolism close to the thermodynamic limits and its implications for digestion applications. Bioresour. Technol. 247, 1095–1106. https://doi.org/10.1016/j.biortech.2017.09.103.

Lim, J.W., Chen, C.-L., Ho, I.J.R., Wang, J.-Y., 2013. Study of microbial community and biodegradation efficiency for single- and two-phase anaerobic co-digestion of brown water and food waste. Bioresour. Technol. 147, 193–201. https://doi.org/10.1016/j.biortech.2013.08.038.

Liu, C.-G., Xue, C., Lin, Y.-H., Bai, F.-W., 2013. Redox potential control and applications in microaerobic and anaerobic fermentations. Biotechnol. Adv. 31, 257–265. https://doi.org/10.1016/j.biotechadv.2012.11.005.

Luo, J., Chen, Y., Feng, L., 2016. Polycyclic aromatic hydrocarbon affects acetic acid production during anaerobic fermentation of waste activated sludge by altering activity and viability of acetogen. Environ. Sci. Technol. 50, 6921–6929. https://doi.org/10.1021/acs.est.6b00003.

Luton, P.E., Wayne, J.M., Sharp, R.J., Riley, P.W., 2002. The mcrA gene as an alternative to 16S rRNA in the phylogenetic analysis of methanogen populations in landfill. Microbiology 148, 3521–3530.

Morris, R.L., Schmidt, T.M., 2013. Shallow breathing: bacterial life at low O2. Nat. Rev. Microbiol. 11, 205–212. https://doi.org/10.1038/nrmicro2970.

Nguyen, D., Khanal, S.K., 2018. A little breath of fresh air into an anaerobic system: how microaeration facilitates anaerobic digestion process. Biotechnol. Adv. 36, 1971–1983. https://doi.org/10.1016/j.biotechadv.2018.08.007.

Pruesse, E., Quast, C., Knittel, K., Fuchs, B.M., Ludwig, W., Peplies, J., Glöckner, F.O., 2007. SILVA: a comprehensive online resource for quality checked and aligned ribosomal RNA sequence data compatible with ARB. Nucleic Acids Res. 35, 7188–7196. https://doi.org/10.1093/nar/gkm864.

Sawatdeenarunat, C., Surendra, K.C., Takara, D., Oechsner, H., Kumar Khanal, S., 2015. Anaerobic digestion of lignocellulosic biomass: challenges and opportunities. Bioresour. Technol. 178, 178–186. https://doi.org/10.1016/j.biortech.2014.09.103.

Schloss, P.D., Westcott, S.L., Ryabin, T., Hall, J.R., Hartmann, M., Hollister, E.B., Lesniewski, R.A., Oakley, B.B., Parks, D.H., Robinson, C.J., Sahl, J.W., Stres, B.,

Thallinger, G.G., Van Horn, D.J., Weber, C.F., 2009. Introducing mothur: open-source, platform-independent, community-supported software for describing and comparing microbial communities. Appl. Environ. Microbiol. 75, 7537–7541. https://doi.org/10.1128/AEM.01541-09.

Steyer, J.P., Bernard, O., Batstone, D.J., Angelidaki, I., 2006. Lessons learnt from 15 years of ICA in anaerobic digesters. Water Sci. Technol. 53, 25–33. https://doi.org/10.2166/wst.2006.107.

Surendra, K.C., Khanal, S.K., 2014. Effects of crop maturity and size reduction on digestibility and methane yield of dedicated energy crop. Bioresour. Technol. 178, 187–193. https://doi.org/10.1016/j.biortech.2014.09.055.

Surendra, K.C., Sawatdeenarunat, C., Shrestha, S., Sung, S., Khanal, S.K., 2015. Anaerobic digestion-based biorefinery for bioenergy and biobased products. Ind. Biotechnol. 11, 103–112. https://doi.org/10.1089/ind.2015.0001.

Thauer, R.K., Jungermann, K., Decker, K., 1977. Energy conservation in chemotrophic anaerobic bacteria. Bacteriol. Rev. 41, 100–180. https://doi.org/10.1073/pnas.0803850105.

Treu, L., Campanaro, S., Kougias, P.G., Zhu, X., Angelidaki, I., 2016. Untangling the effect of fatty acid addition at species level revealed different transcriptional responses of the biogas microbial community members. Environ. Sci. Technol. 50, 6079–6090. https://doi.org/10.1021/acs.est.6b00296.

Ward, A.J., Hobbs, P.J., Holliman, P.J., Jones, D.L., 2008. Optimisation of the anaerobic digestion of agricultural resources. Bioresour. Technol. 99, 7928–7940. https://doi.org/10.1016/j.biortech.2008.02.044.

Yuan, X., Shi, X., Zhang, D., Qiu, Y., Guo, R., Wang, L., 2011. Biogas production and microcystin biodegradation in anaerobic digestion of blue algae. Energy Environ. Sci. 4, 1511. https://doi.org/10.1039/c0ee00452a.

Zhu, M., Lü, F., Hao, L.-P., He, P.-J., Shao, L.-M., 2009. Regulating the hydrolysis of organic wastes by micro-aeration and effluent recirculation. Waste Manag. 29, 2042–2050. https://doi.org/10.1016/j.wasman.2008.12.023.